



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Binary Search- Iterative

BINARY SEARCH- ITERATIVE

Read n daily temperatures into an array, sort ascending using any method. Use binary search (iterative) to find first day $\geq 30^{\circ}\text{C}$ and last day $\leq 25^{\circ}\text{C}$. Display temperatures, indices (0-based), or ""Not found"". Validate sorted; $n \leq 50$.

Sample Input: 6

22 28 31 24 29 32

Sample Output (after sort):

22 24 28 29 31 32

First ≥ 30 : index 4 (31)

Last ≤ 25 : index 1 (24)

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int binaryFirstGE30(int arr[], int n) {
4     int low = 0, high = n - 1;
5     int ans = -1;
6
7     while (low <= high) {
8         int mid = low + (high - low) / 2;
9         if (arr[mid] >= 30) {
10             ans = mid;
11             high = mid - 1;
12         } else {
13             low = mid + 1;
14         }
15     }
16     return ans;
17 }
18
19 int binaryLastLE25(int arr[], int n) {
20     int low = 0, high = n - 1;
21     int ans = -1;
```

OUTPUT

```
22 24 28 29 31 32
First >=30: index 4 (31)
Last <=25: index 1 (24)
```

INPUT

6 22 28 31 24 29 32

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